Understanding Image Limiting Magnitude

A beginner's guide to CCD images, photometry, and the J-POD implementation

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How to use this guide

This guide explains how an astronomical image can be assigned a limiting magnitude, how J-POD currently estimates one for V-band images, and what must be verified before that estimate can be considered robust across many images. It is intended as a reference: terms are introduced before they are used, and the current implementation is distinguished from future work.

The short version is:

> A limiting magnitude is not a property written into an image by nature. It is an estimate based on a stated filter, calibration, noise model, aperture, and detection threshold.

J-POD's current automatic estimate is an **image-wide, point-source, V-band estimate using sky Poisson noise only**. It is an initial model, not yet a validated measurement of completeness. Results should be treated cautiously until image units, zero-point conventions, calibration scatter, and detection performance have been checked.

Status labels used here

- **Implemented** means the behavior exists in the current J-POD source.
- **Needs verification** means code exists, but an assumption or accuracy has not yet been established across different data sets.
- **Planned** means it is a recommended next development step, not current behavior.

The basic idea

Astronomers often express the brightness of a source using a **magnitude**. Magnitudes run in the opposite direction from everyday brightness numbers: smaller magnitude means brighter light. A difference of five magnitudes means a factor of 100 in brightness; one magnitude is a factor of about 2.512.

An image's **limiting magnitude**, written here as V_{lim} for the V filter, is the approximate magnitude of the faintest point-like source that the image could detect under a specified rule. For example, an SNR-3 limit means the limiting source is estimated to produce a signal three times the estimated noise in the chosen measurement aperture.

This does not mean every source brighter than that value is guaranteed to be found, or every fainter source is guaranteed to be missed. Real detection depends on the detector, image processing, source shape, background, and detection software. A calculated limit is best understood as a model-based threshold; injection-and-recovery tests are a stronger way to measure actual detection completeness.

Why J-POD needs a limit

J-POD compares predicted asteroid magnitudes with an image's sensitivity. Its magnitude probability is currently calculated as

$$P_{\text{mag}} = \Phi\left(\frac{V_{\text{lim}} - V_{\text{trailing}}}{\sigma_V}\right),$$

where:

- V_{trailing} is the asteroid's predicted V magnitude adjusted for estimated trailing loss;
- V_{lim} is the image-wide limit;
- σ_V is the adopted uncertainty in the asteroid magnitude; and
- Φ is the cumulative distribution function of a normal distribution.

This gives one probability based on magnitude. J-POD also calculates a positional probability and combines the two. At present, V_{lim} is treated as exact in this probability calculation. Uncertainty in the limit is not yet propagated into P_{mag}.

What is inside a CCD image?

Light becomes charge, then numbers

A digital astronomical camera has light-sensitive detector elements called **pixels**. A photon that is absorbed by the detector can release an electron. The

detector collects these electrons as electrical charge during an exposure. Electronics then measure the charge and an analog-to-digital converter turns the measurement into a number stored in the image.

The chain is approximately:

```
photons -> detector electrons -> measured voltage/charge -> digital pixel value
```

The number in a FITS image is therefore not automatically a photon count or an electron count. It may be a digital number, a calibrated electron value, a rate, or a processed value. FITS headers sometimes describe the units, but headers are not always complete or consistent.

Electrons, ADU, DN, and gain

- **Electron (e-)**: a unit of detector charge. Photon arrival is statistical; for N detected photons/electrons, the Poisson standard deviation is approximately $\sqrt{N}$ electrons.
- **ADU**: analog-to-digital unit, a digital number representing the detector measurement.
- **DN**: digital number. In many image files it is used much like ADU.
- **Gain**: a conversion factor between image units and electrons. A common definition is $g = \text{electrons}/\text{ADU}$. A gain of $0.83\ \text{e}^-/\text{ADU}$ means one ADU corresponds on average to 0.83 collected electrons.
- **BUNIT**: a FITS header keyword that may state the units of pixel values. It can be absent, generic, or insufficient to describe whether values are integrated over the exposure or already divided by exposure time.

If pixel values are in ADU, Poisson noise is not generally $\sqrt{\text{ADU}}$ unless the gain is exactly one electron per ADU. If pixel values are in electrons, gain conversion is not needed for the Poisson-noise calculation. If values are rates, the exposure-time conversion must be handled differently.

Other detector and image effects

These are common terms in CCD/CMOS imaging. They are **not all included** in J-POD's current limiting-magnitude estimate.

- **Bias**: an electronic baseline added to avoid negative digitized values.
- **Dark current**: electrons produced by the detector even without incoming light. It generally increases with exposure time and temperature.
- **Read noise**: uncertainty added when the detector electronics read each pixel.
- **Flat field**: a correction for pixel-to-pixel sensitivity differences and illumination effects such as vignetting.
- **Stacking**: combining multiple exposures. Noise depends on how they were combined, rejected, rescaled, and weighted.

- **Saturation**: the detector or digitizer reaches its maximum representable value. Saturated stars cannot provide reliable flux measurements.

Images may already have had bias, dark, and flat corrections applied. Reduction software may also rescale pixel values. Therefore, a FITS file cannot always be interpreted from its pixel values alone; image provenance and header documentation matter.

Filters, bands, and magnitudes

A filter selects colors of light

A filter passes some wavelengths and blocks others. Astronomers name standardized wavelength ranges with band labels such as **B**, **V**, and **R**. The V band is centered in visible light; R is redder. Other systems, such as the Sloan survey, use labels such as g', r', and i'.

The filter's full wavelength response is its **passband** or **transmission curve**. A filter labelled `R` is not guaranteed to be identical to every other filter labelled `R`: detector sensitivity, filter glass, optics, and standard-system definitions can differ.

A magnitude always belongs to a band and a system

A magnitude is measured through a passband. The same star can have different values in V and R because its emitted light varies with wavelength. A catalog value of $V = 15$ is not automatically an R magnitude of 15.

Magnitude systems also define how flux is referenced. Two common examples are the Vega system and the AB system. A value in one system should not be mixed with another without the appropriate convention or transformation.

Color indexes and asteroid comparisons

A **color index** is a difference between magnitudes in two filters, for example

$$V - R.$$

It describes how an object's measured brightness changes between those bands. A redder object and a bluer object with the same R magnitude can have different V magnitudes.

For an R-filter image, a defensible workflow would be:

1. Calibrate the image in its actual R-like passband using compatible reference-star magnitudes.
2. Compute the image's limit in that native passband, such as R_{lim}.

3. If comparison with asteroid V magnitudes is required, convert the one image-wide limit using an explicitly chosen representative asteroid color:

$$V_{\text{lim}} \approx R_{\text{lim}} + (V - R)_{\text{assumed}}.$$

The assumed color would be common to the image and the asteroid population being studied, so the result remains one image-wide number. It is approximate: individual asteroids have different colors. J-POD does **not** implement this conversion yet.

Current filter scope

Implemented: catalog-derived zero-point calibration is restricted to images explicitly identified as V band because the local ASTAP V50 catalog is being used for Johnson V magnitudes.

Not implemented: calibration in R or Sloan bands, filter transformations, or asteroid-color corrections. Non-V images continue to use the user-provided `--vlim` fallback. A header zero point can still be read, but its filter and photometric convention are not independently validated.

Star images, FWHM, and apertures

Why a star is a spot, not one pixel

Even a point-like star is recorded across multiple pixels. The telescope, atmosphere, focus, tracking, and detector spread its light into a small image called the **point-spread function (PSF)**. A star's apparent width can vary across the image and between exposures.

FWHM

FWHM means full width at half maximum. Imagine a cross-section through a star's central peak. FWHM is the width of the star at the points where its brightness has fallen to half the peak height. It is a width, not a radius.

J-POD estimates the widths of suitable stellar profiles and uses their robust median. In the current measurement pass it seeks compact, relatively round, bright profiles; excludes saturated and nearby blended candidates; and uses at most the brightest 100 profile candidates. A plausible FITS header FWHM is preferred. If the header value differs from the measured value by more than a factor of two, J-POD reports a note and uses the image-measured value.

Aperture and aperture flux

An **aperture** is a region around a source whose pixels are summed to estimate its brightness. J-POD currently uses a circular aperture with **radius equal to**

one FWHM. For FWHM $= 3.28$ pixels, this means radius 3.28 pixels, not diameter 3.28 pixels.

The **aperture flux** is the sum of the pixel values inside that circle after subtracting the estimated background. It is measured in the image's pixel units. It is not the brightness of one pixel.

For a circular aperture of radius $r =$ FWHM, the continuous-area approximation is

$$A_{\text{pix}} \approx \pi \, \text{FWHM}_{\text{pixels}}^2.$$

The real software sums discrete pixels whose centers fall inside the circle; the limiting-magnitude calculation uses the continuous area approximation above. For an ideal two-dimensional Gaussian PSF, a radius of one FWHM encloses about 94% of the total light. Real profiles may differ.

The catalog zero point is derived using this same aperture. That makes the zero point tied to the aperture measurement. A future validation should check whether this aperture is appropriate and whether an aperture correction is needed for different PSFs or image-processing pipelines.

Background and noise

What is the background?

The image **background** is the broad level underneath sources. It can include sky brightness (natural airglow, moonlight, light pollution, and scattered light), detector offset, dark signal, and residuals from image processing. Thus "measured image background" is not necessarily pure astronomical sky.

Background and background noise are different:

- **Background level** is the average/typical pixel value underlying sources.
- **Noise** is the random variation around that level.

A bright, smooth sky can have a high background and correspondingly high photon noise. A low background does not guarantee low noise if read noise or processing artifacts are large.

How J-POD estimates it

Implemented: J-POD uses Photutils `Background2D` with a 50-by-50-pixel mesh and a median estimator to make a spatial background map. It subtracts this map to find stars and estimate the residual scatter used by star detection.

For the limiting-magnitude formula, J-POD takes the median of the background map. It treats this as the typical image-wide background per pixel

for the exposure and uses that value directly in the sky-noise formula. A background rate (background divided by exposure time) can help compare images with different exposure durations, but is not needed for this per-image calculation. The residual-noise estimate used during star finding is **not** the σ_{sky} in the current limiting-magnitude equation.

Using one median creates one image-wide limit. It does not describe local changes due to vignetting, gradients, crowded fields, nebulosity, or a bright Moon gradient.

Sky Poisson noise

Photons arrive randomly. If the expected number of detected sky electrons is N_e, the Poisson standard deviation is approximately $\sqrt{N_e}$ electrons and the variance is N_e electrons squared.

Let:

- A_{pix} be the number of pixels in the measurement aperture;
- B_{image} be the accumulated background in image units per pixel for the whole exposure;
- t_{exp} be exposure time in seconds; and
- g be gain in electrons per image unit.

The current implementation uses the median full-exposure background B_{image} directly. It estimates the sky-noise variance in image units squared as

$$\sigma^2_{\text{sky,image}} = A_{\text{pix}}\, B_{\text{image}}/g.$$

Then

$$\sigma_{\text{sky,image}} = \sqrt{A_{\text{pix}}\, B_{\text{image}}/g}.$$

If pixel units are electrons, use the equivalent formula with $g = 1$ electron per image unit. If pixel units are ADU, dividing the variance by electrons-per-ADU converts the electron shot noise back to ADU units.

Important current assumption: J-POD reads `EGAIN` first, then `GAIN`, and interprets either as electrons per image unit. It does not currently inspect `BUNIT` to determine whether pixel values are ADU, electrons, or rates. If no gain is found, it assumes $g = 1$. These assumptions are not safe for every FITS file and are a high-priority item to improve.

Noise not yet included

The current limit includes sky Poisson noise only. It does not include:

- read noise from reading the detector;
- dark-current shot noise;
- source Poisson noise at the detection threshold;
- uncertainty in estimating/subtracting the background;
- correlated noise introduced by resampling or stacking;
- cosmic rays, bad pixels, or other defects.

Moonlight is not a separate noise term in this first model; its smooth contribution may raise the measured background, but gradients may make a single median inadequate. The chosen omissions mean the current limit is an approximation, not a complete detector noise model.

Zero point: connecting image numbers to magnitudes

What a zero point does

The image records flux in instrumental units, not standard magnitudes. A **photometric zero point** is a calibration constant that relates measured flux to a catalog magnitude.

J-POD uses a count-rate convention:

$$m = ZP - 2.5 \log_{10}(F/t_{\text{exp}}),$$

or equivalently

$$ZP = m + 2.5 \log_{10}(F/t_{\text{exp}}),$$

where F is the background-subtracted aperture flux in image units and F/t_{exp} is its rate in image units per second. Brighter sources have larger flux and smaller magnitudes.

This convention means the numerical zero point depends on the units and exposure-time convention of the flux. A zero point for ADU/s is not automatically interchangeable with a zero point for electrons/s or for uncorrected ADU totals.

Current zero-point procedure

For each image, the current pipeline:

1. Loads pixels, WCS, exposure, filter, and selected header metadata.

2. Measures background, FWHM, candidate star profiles, and aperture fluxes in one shared image-analysis pass. If there are no suitable stars and no usable header FWHM, it preserves the background measurements and marks FWHM unavailable instead of stopping.
3. Checks for a numeric `MAGZERO` header value; if present, uses it. Otherwise it checks `PHOTZP`. If both are present but differ, `MAGZERO` wins and the program warns.
4. If there is no header zero point, only proceeds with catalog calibration if the filter is recognized as V and the local V50 catalog is available.
5. Selects nearby V50 sky-cell files, projects catalog coordinates into image pixels using the FITS WCS, and matches catalog stars to measured stars.
6. Requires at least five matches, uses at most the 50 brightest measured stars, rejects strong outliers with a median/MAD rule, and reports the median of individual zero-point estimates.

If a usable header zero point exists, the code returns it without opening the catalog. The shared image-measurement pass has already run, however, because background measurements are needed by the pipeline.

When FWHM cannot be measured and is also absent from the header, J-POD skips the trailing-loss correction and the automatic limiting-magnitude estimate. It retains the unadjusted asteroid V for magnitude probabilities and keeps the user-provided `--vlim`. The output marks `V_trailing`, the trail-to-FWHM ratio, and trailing loss as unavailable.

What the J75 numbers mean

The test image `data/J75_01_20110720204510.fits` has `FILTER='V'`, `EXPTIME=90 s`, and header gain `EGAIN=0.83 electrons/ADU`. The current processing produced approximately:

- image-measured FWHM: 3.28 pixels (the header says 90 pixels, so the image-measured value is used);
- matched catalog stars: 7;
- median catalog-derived zero point: 21.713;
- robust scatter of individual star zero points: 0.603 mag;
- median background: 2491.7 image units/pixel for the exposure, or 27.69 image units/pixel/second;
- approximate aperture area: $\pi(3.28)^2 \approx 33.89$ pixels;
- sky noise in the aperture: about 318.95 image units; and
- calculated sky-only limit at SNR 3: about $V_{\mathrm{lim}} = 19.15$.

The 0.603-mag scatter is a robust spread among the individual matched-star zero-point estimates. It is **not yet a confidence interval or standard error on** V_{lim}. It is relatively large and makes this J75 estimate provisional. More and better matches, a validated photometric aperture, and understanding of the header/calibration conventions are needed before relying on it for quantitative probabilities.

Header zero points need validation

The current code accepts finite `MAGZERO` and `PHOTZP` values without independently establishing their filter, flux units, aperture convention, or whether exposure time is already included. A header value can therefore be numerically usable but physically incompatible with the image measurements. Checking its plausibility against catalog stars is a useful future feature, but that comparison must first ensure both values use the same filter and flux convention.

The current limiting-magnitude calculation

Formula

With aperture radius equal to FWHM,

$$A_{\mathrm{pix}} \approx \pi\,\mathrm{FWHM}_{\mathrm{pixels}}^{2}.$$

The program estimates sky noise as

$$\sigma_{\mathrm{sky}} = \sqrt{A_{\mathrm{pix}}\,B_{\mathrm{rate}}\,t_{\mathrm{exp}}/g}.$$

It then estimates the limiting magnitude with

$$V_{\mathrm{lim}} = ZP + 2.5\log_{10}\left(\frac{t_{\mathrm{exp}}}{\mathrm{SNR}_{\mathrm{threshold}}\,\sigma_{\mathrm{sky}}}\right).$$

Here ZP is calibrated for image units per second and σ_{sky} is in image units. In this shorthand, the zero point defines the reference rate unit. In code, the equivalent operation is to calculate the limiting aperture-flux rate, $\mathrm{SNR}_{\mathrm{threshold}}\,\sigma_{\mathrm{sky}}/t_{\mathrm{exp}}$, and compare it with the calibrated rate zero point.

SNR threshold

SNR is signal-to-noise ratio: source signal divided by the uncertainty in that signal measurement. The default threshold is 3. It can be changed with `--snr`, for example `--snr 5`. A higher threshold is more conservative and produces a brighter (numerically smaller) limiting magnitude.

How the limit is used

For V-band images with an available zero point and usable background, the calculated value replaces `--vlim` for the detection-probability calculation and plots. If the image is not recognized as V, the zero point is unavailable, or the

limiting calculation fails, J-POD falls back to the supplied `--vlim` value (default 21.5).

The calculation produces one global point-source limit using the median background and one representative FWHM. Asteroid trails are handled separately by the existing trailing-loss estimate, which adjusts predicted asteroid magnitude before computing the magnitude probability. This does not make the image limit a local map of sensitivity.

What has been implemented, and what has not

Implemented now

- FITS loading collects filter, exposure, selected zero-point and FWHM keywords, saturation level, and gain keywords.
- One shared image-analysis pass estimates a spatial background map, a robust residual scatter for star detection, stellar profiles, FWHM, and aperture fluxes.
- Header `MAGZERO` is preferred to `PHOTZP`; if neither is present, V50 catalog matching is attempted only for recognized V filters.
- V50 lookup reads sky cells near measured stars rather than scanning the whole catalog.
- Catalog-derived calibration requires at least five matches and reports robust scatter.
- A sky-only V-band limit is calculated using the median background, FWHM-radius aperture, gain if present (otherwise assumed unit gain), and configurable SNR threshold.
- The automatic limit is passed into magnitude-probability calculations and plots; `--vlim` remains a fallback.

Explicitly not implemented

- General recognition of whether pixel values are ADU, electrons, counts per second, or rescaled/stacked units.
- Validation that header zero points share the filter and rate/aperture convention used by the calculation.
- Read-noise, dark-current, source-shot-noise, or resampling-noise terms.
- A spatial limiting-magnitude map or a validated typical/conservative image-wide background policy.
- Uncertainty propagation from FWHM, zero point, background, gain, or catalog magnitudes into a V_{lim} interval.
- Propagation of limit uncertainty into a range or distribution for P_{mag}.
- Calibration for non-V filters or a justified conversion from another band to asteroid V magnitudes.

- Validation against measured detection completeness using artificial-source injection and recovery.

Remaining work, in priority order

This is the recommended sequence for making the estimate robust across many images. Earlier steps support the correctness of later ones.

Priority 1 — Establish pixel units and gain

Read and interpret `BUNIT`, `EGAIN`/`GAIN`, comments, and image provenance. Recognize integrated ADU, integrated electrons, rates, and known processed products where possible. Convert sky Poisson variance into the same units as the flux used for the zero point. If units cannot be determined, do not silently assume unit gain: report the assumption or fall back safely.

Priority 2 — Verify the zero point and J75 calibration scatter

Inspect the individual matched stars and their positions, fluxes, magnitudes, and per-star zero points. Investigate J75's 0.603-mag robust scatter. Check match quality, saturation, blending, background subtraction, catalog uncertainty, filter identity, and whether the aperture/zero-point convention is valid. Define and validate the meanings of header zero-point keywords.

Priority 3 — Define the image-wide background and aperture rule

Decide whether the limit means typical sensitivity, a conservative value, or another chosen statistic. Test the median background against gradients, vignetting, bright stars, bad pixels, crowded fields, and source-free areas. Validate whether one FWHM is the right aperture radius and how its discrete pixel area should be calculated. Document the point-source assumption and any aperture correction.

Priority 4 — Propagate uncertainty into the limit and probability

Estimate uncertainty in FWHM, zero point, and background. Because FWHM affects both the calibration aperture and the sky-noise aperture, propagate these inputs together, for example by resampling the calibration stars and stellar-profile measurements and recalculating ZP, σ_{sky}, and V_{lim}. Use the resulting limit distribution to compute a probability interval or distribution

for P_{mag}. Keep asteroid magnitude uncertainty separate from image-limit uncertainty, and state which sources are included.

Priority 5 — Add other noise sources

After the sky-only path is validated, add read noise, dark-current noise, and source shot noise when the image metadata supports them. Account for correlated noise in resampled/stacked data where relevant. These terms should be added in variance space with compatible units, not by adding standard deviations blindly.

Priority 6 — Expand filter support

Identify the actual filter system, use compatible reference-star magnitudes, and calibrate limits in the native band. If asteroid V magnitudes remain the comparison target, provide an explicit color-based conversion with a documented representative asteroid color and uncertainty. Do not silently treat R, Sloan r', and V as interchangeable.

Priority 7 — Validate performance on known and synthetic cases

Use synthetic images with known backgrounds, gain, FWHM, zero point, and source fluxes. Test real images with trustworthy photometry. Check sensible responses to changes in exposure, background, and seeing. Finally, inject artificial sources and measure recovery rate versus magnitude; this tests actual detection completeness rather than only the analytic noise threshold.

Immediate next development task

The most useful next task is **Priority 1 together with the J75 checks in Priority 2**: establish image units and gain handling, then inspect why the seven matched J75 stars produce a large zero-point spread. Until those checks are done, the automatic number is a useful experiment but not a robust cross-instrument result.

Glossary

ADU (analog-to-digital unit) — A digital value representing measured detector charge or voltage. Often similar to DN.

Airmass — A measure of the amount of atmosphere along a line of sight. Larger airmass generally means more extinction. A calibrated in-field zero point already reflects the observed atmosphere for that exposure, so an additional correction must not be applied without checking its convention.

Aperture — A selected region around a source whose pixel values are summed.

Aperture flux — Background-subtracted sum of image values in an aperture.

Aperture correction — Adjustment for light outside the chosen aperture when converting aperture flux to a total-source flux convention.

Astrometric catalog — A catalog giving sky positions; it may or may not also provide suitable photometric magnitudes.

Background — The image level underneath sources; it may combine sky, detector, and processing contributions.

Band / passband — A wavelength range selected by a filter and its detector/optical response, such as V or R.

Bias — Electronic offset added to detector measurements.

CCD / CMOS — Common solid-state detector technologies used in digital cameras. Both convert incoming light into measurable charge.

Color / color index — Difference between magnitudes in different bands, such as $V - R$; describes how an object's brightness changes with wavelength.

Dark current — Detector-generated charge accumulated without incident light.

DN (digital number) — A pixel's stored numerical value; often used interchangeably with ADU.

Exposure time (t_{exp}) — Duration for which the detector collected light.

FITS — Flexible Image Transport System, a standard astronomical file format containing arrays and metadata headers.

FWHM — Full width at half maximum; a measure of a star's image width.

Gain (g) — Conversion between image units and electrons, commonly electrons per ADU.

Limiting magnitude (V_{lim}) — Estimated faintest source magnitude detectable under a stated SNR and measurement procedure.

Magnitude — Logarithmic brightness scale; a smaller value is brighter. Magnitudes must be associated with a band and photometric system.

MAD (median absolute deviation) — Robust measure of spread around a median. A scaled MAD is used in the code to reduce the influence of outliers.

Photometry — Measurement and calibration of source brightness.

Photon shot noise / Poisson noise — Random variation in photon arrivals; for N counts the standard deviation is approximately $\sqrt{N}$.

PSF (point-spread function) — Image shape produced by a point-like source such as a distant star.

Read noise — Noise introduced by detector readout electronics.

SNR (signal-to-noise ratio) — Measured source signal divided by its estimated uncertainty.

Zero point (ZP) — Calibration constant relating instrumental flux/rate to a standard magnitude for a specified filter and convention.

WCS (World Coordinate System) — FITS metadata mapping image pixels to sky coordinates such as right ascension and declination.

Further reading

These sources provide additional detail and define relevant data standards:

1. **J-POD implementation:** `jpod/image_geometry.py`, `jpod/image_analysis.py`, `jpod/photometric_calibration.py`, `jpod/detection.py`, `jpod/trailing_loss.py`, and `docs/pipeline_guide.md`.
2. **FITS Standard:** NASA/GSFC FITS Support Office, https://fits.gsfc.nasa.gov/fits_standard.html.
3. **Photutils background estimation:** Astropy Photutils documentation, https://docs.astropy.org/projects/photutils/en/stable/user_guide/background.html.
4. **ASTAP star databases:** ASTAP documentation and database downloads, https://www.hnsky.org/astap.htm#index.
5. **APASS multi-band photometry:** AAVSO APASS overview, https://archive.aavso.org/apass.
6. **Asteroid multi-band photometry:** NASA Planetary Data System Small Bodies Node, SDSS Moving Object Catalog, https://sbn.psi.edu/pds/resource/sdssmoc.html.

The code behavior described here was checked against the J-POD workspace on 9 November 2026. External catalog products and documentation can change; check their current release notes before relying on catalog details.